# Museum without walls: a digital site-specific museum in São Miguel das Missões

Karolina Rodrigues Ziulkoski, Bolota Exhibition Design, Brazil

*http://www.karolinaziulkoski.com/Museum-WIthout-Walls*

*Abstract: Museum Without Walls is a model for site-specific museums. It provides immersive experiences with no physical interventions: the content is all digital and triggered on site only, bringing information and meaning to historical sites.*

Keywords: augmented reality, projection mapping, digital museum, transmedia storytelling

## The project

Must museums have rooms and walls? From this question came the concept of “Museum Without Walls”, a model for heritage areas. The museum exists entirely in the digital realm, and is experienced on site only. History is told in parts using different technologies - each tells the tale most suitable to its characteristics. For this proposal the case study is São Miguel das Missões, Brazil, a former Jesuit mission, and the chosen media are video mapping and Augmented Reality. The daily projection serves as a collective night time experience that depicts the epic final battle of the mission. The Augmented Reality application allows for deeper individual exploration, intimately displaying daily life in the mission.

## Research

“Exhibits not only provide an opportunity for visitors to look but to think, to explore, to wonder, and to investigate” (Olofsson 1979, p.166). This project aims at educating the visitors of São Miguel das Missões about life in the mission, how it was during the pinnacle of the reduction and what led to its demise. Moreover, its goal is to do so through methods that are fun and engaging.

Based on Levi-Strauss’s magical thought, visitors learn by context, association and previous experiences, as opposed to a formal education model (quoted in Graburn p.179). Indeed, for Falk and Dierking, the museum visit is an interchange between three contexts: personal, social and physical – visitors continuously create each context, and the interaction among these result in their experience (1992, p.2). Research “suggests that the experience of exhibits is fundamentally produced in social interaction between visitors” (Lehn p.1353). Most people choose to visit a museum in groups, so a large part of their attention is devoted to their social peers (Falk and Dierking 1992, p.41). Interaction among a group, and even with other visitors, shapes the way in which people understand an exhibition. Social interactions play such a large role in a museum visit that they are rarely forgotten, and, sometimes, they are the most recalled factor many years after the exhibition experience (Falk and Dierking 1992, p.54). A museum experience is shaped by the social context – interpretation and learning outcomes are very much the results of a group outcome (Falk and Dierking 1992, p.54).

This research informed the products chosen to compose the case study for Museum Without Walls, as well as visitor’s behavior and current events on site. An existing daily light and sound show has been showcased for thirty years and features notable actors, such as Fernanda Montenegro, the only Brazilian to have been nominated for an acting Oscar. Therefore, it is reasonable to renew the spectacle and incorporate it as one of the museum's features. The show is already known by visitors as something they should attend to during their visit, and is a highly social and communal experience. However, at their visit to the site during daytime, there's no aid. School groups have guides, but everyone else is essentially alone and there's no information, or even labels. This provides an opportunity for a second component for the museum. Considering the literature review above, and site observation which points to the fact that the audiences visit the site in small groups, there should be a product to allow for education about the mission in a fun way, and in a way that connects to the intimate context of these small group visits. People already have highly technological devices, such as smartphones. To use the visitor's own device is not only more practical, but also reduces costs and gives the visitors themselves more freedom - it's their device, they can use it at their own discretion.

The technology chosen to be used in a mobile and tablet application for use in site is Augmented Reality. According to Kipper and Rampolla "Augmented Reality is taking digital or computer generated information, whether it be images, audio, video, and touch or haptic sensations and overlaying them in a real-time environment" (2012, p.1). Essentially, it allows the user to see the real world in their screen with the computer generated content on top of it. In this particular case, it can provide a view of how the mission was before becoming a ruin. The Augmented Reality technology is attached to a context: it works by providing an image it recognizes or through GPS coordinates. So the application will only deliver content when inside the site, serving as an aid and not an alternative to a visit to the place. Although mobiles and tablets are being used more and more as tools for individualization, many applications can be used in groups. Parents use smartphones with their children. One tablet can serve an entire group at the same time in many museums' educational programs. As long as the application is built with interaction between small groups of people in mind, it is absolutely feasible for groups of up to four people to share it, making it a social experience.

## Audience

"People are the only reason for museums to exist" (Dean 1996, p.19). Defining who those people are for a given exhibition is a key factor. For this project, the targeted public represents a very broad group - men and women of all ages. Three groups are recurring visitors: school classes, elderly visitors and couples, with or without kids.

## User scenario

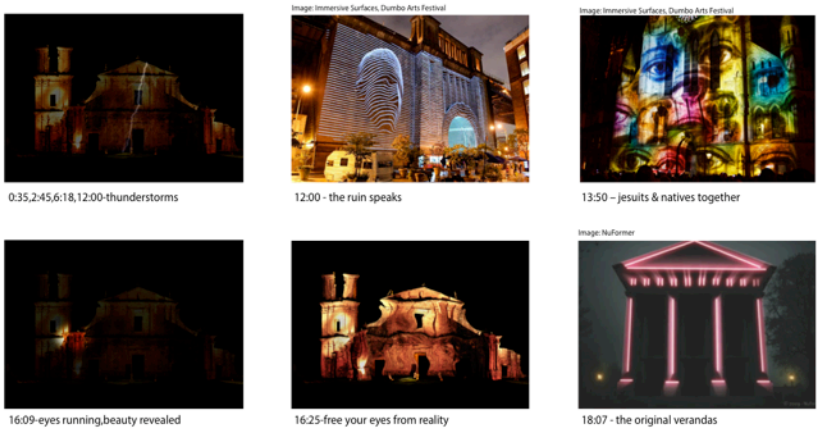
During the day, visitors use their smartphones to visit the site at their own pace, with freedom to choose between the Augmented Reality visit and the audio guides. It can be shared in a small group, so there is no need for each person to have their own device. The application is meant for use on site only – if users delete it later, it's not an issue. At night, they may come back to the site – or just remain – for the projection mapping show.

## Implementation

With the products that comprise the museum already settled - the existing light and sound show and the idea of an Augmented Reality application – the project developed in a very straightforward way. For the light and sound show the best option available was projection mapping. This technique turns objects into display surfaces for video projection by spatially mapping them. It allows for the use of the existing script and sound narrative, updating them with visuals by using the church ruins as a canvas. The most important issue here is content. The majority of projection mapping events around the world – developed so far – display only visuals with no narrative. For this project, the process happened in the reverse order: the sound narrative guided the visuals, complementing them and delivering meaning. The 45-minute script tells the history of the epic battle which led to the end of the mission. It is dense, abstract and very theatrical. Therefore, the visuals do not need to be literal and there is artistic freedom. For this technique, its limitation is also its biggest strength: the manipulation of architecture generating optical illusions. In a long narrative such as the one presented here, there are moments in which the manipulation of architecture fits perfectly; at others, however, architecture has to be completely dismissed.

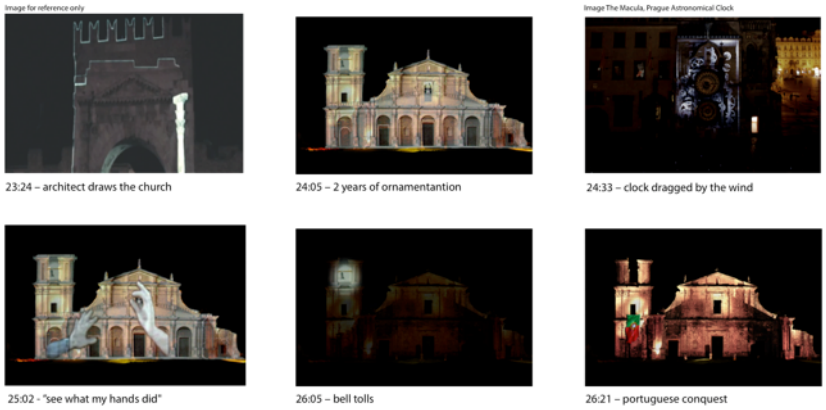
The spectacle was produced and performed in a small scale acrylic laser cut model and documented, as pictured below, using the MadMapper software and a small projector. The storyboard displays key moments in the show:

Image 1: Storyboard 1




*Source(s): Author*

Image 2: Storyboard 2



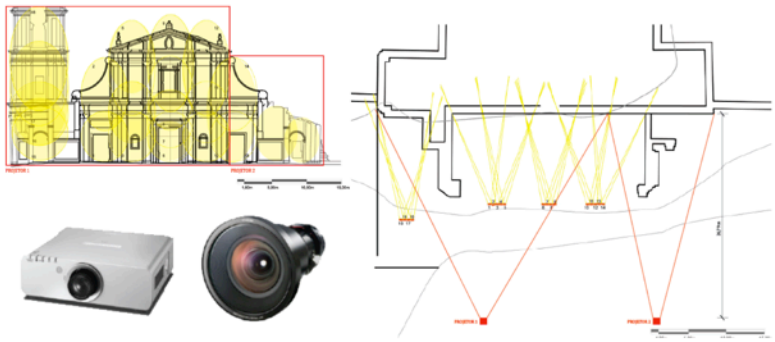

*Source(s): Author*

Image 3: Model Mapping



*Source(s): Author*

Image 4: Projectors and lenses for permanent placement on site



*Source(s): Author*

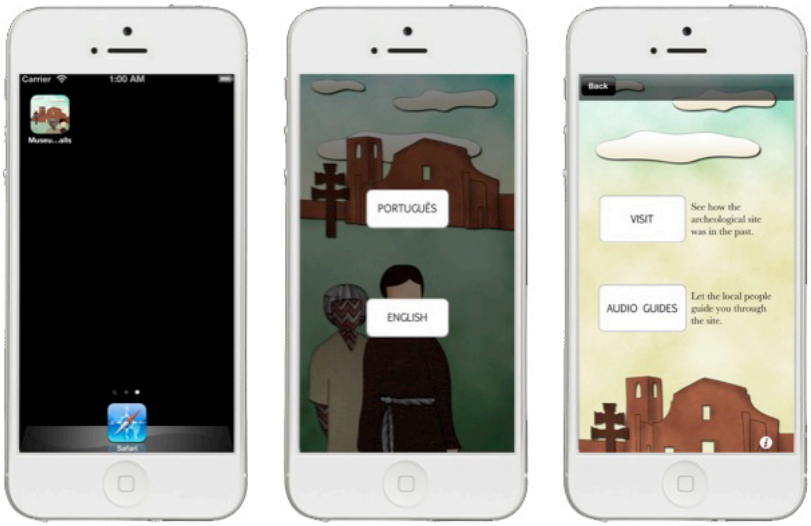
The application development started with research about Augmented Reality providers. The chosen platform was Metaio, since its functionalities are the most suitable to the project. Using GPS

coordinates, it displays the original settlement with an accompanying narration explaining the daily life at the mission. The projection mapping event deals with an epic story and is a spectacle designed for a large audience. In contrast to this, the application is designed for small groups with small devices owned by themselves and allows for a more intimate approach, suitable for chronicles of everyday life. For the content, interviews were conducted with Prof. Nadir Daminani, head of the Center for Missionary Culture at URI, the local university. AR is a powerful tool, but one of its limitations is that it cannot support complex 3D models. However, there are considerable investments being made in this technology, which should improve this situation in a short time.

This is the first part of the application. There is another one, which displays interesting stories from the community. For this, interviews were conducted with local people. The stories they shared were incredible. However, early on, there was a preoccupation regarding how to deliver these narratives, as not everyone is a great storyteller. So the decision was made to produce these stories with actors and sound effects in audio only, so that visitors feel they are being guided through the site, as if the story being told is actually happening around them.

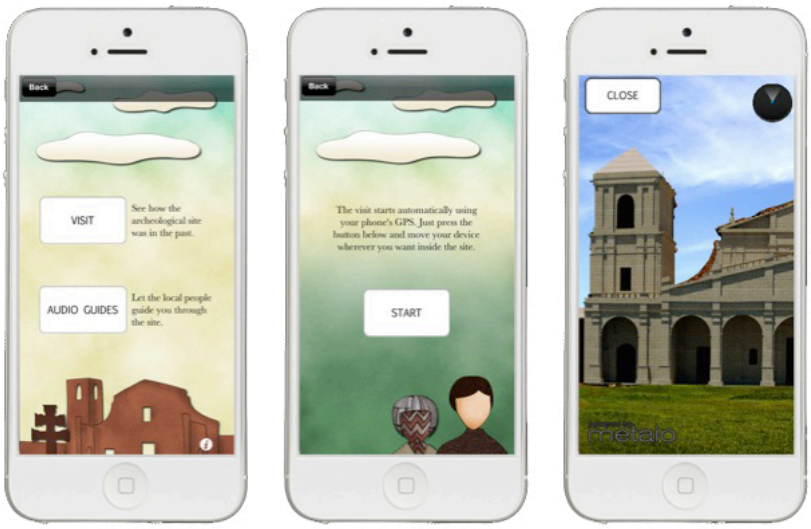
The user experience design of the application is very simple: the first screen offers the option to choose between the audio guides or the Augmented Reality visit. The design is done entirely with illustrations. The color palette displays earthy tones, and the use of textures in unusual places - like the use of a fabric pattern as hair - deliberately steers away from a children's illustration, although they are not completely unappealing to them, as children are a significant part of the audience as well. The application is fully functioning for iOS 6 and up.

Image 5: Application Flow and Design 1




*Source(s): Author*

Image 6: Application Flow and Design 2



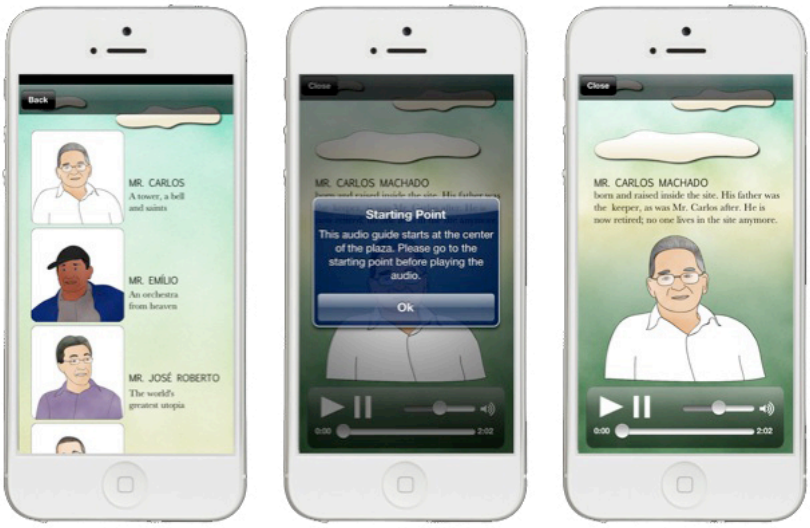

*Source(s): Author*

Image 7: Application Flow and Design 3




*Source(s): Author*

Image 8: Application Flow and Design 4




*Source(s): Author*

All of the design components from both projection mapping and the application are thought to be complementary and consistent, so that the audience understands they are part of a single project. These include illustrations, colours, textures, discourse tone and style, and video visuals.

## Conclusions

This proposal represents new possibilities for museums inside historical sites – it is a model adaptable to the stories each place might have, allowing visitors to learn the history of a place without needing to visit a site and then a museum about the site. More than a technology, it is a concept. Technology can be replaced at any time; what is relevant is the way content is being delivered without interventions.

But most importantly, this project represents a concept of a museum without physical boundaries. Instead of having small objects inside glass boxes, the contents of the museum are stories heard within their original contexts. Visitors are free to move within the site and visit it in their own pace. This project was developed through the lens of the visitor experience and through an analysis of how audiences behave in an exhibition setting, resulting in a museum that will enhance their experience.

## Acknowledgements

Professor Despina Papadoulos, my master's thesis advisor at ITP/ NYU; Professor Nancy Hechinger from ITP/ NYU; Professor Rosana Flouty from Museum Studies/ NYU; Professor Nadir Damiani, from URI; Mr. Dulphe Pinheiro Machado; all intervieews who took some of their time to help in the project.

## References

Dean, D. (1996). *Museum Exhibition Theory and Practice*. London: Routledge.

Falk, J. H., & Dierking, L. D. (1992). *The Museum Experience*. Washington, D.C.: Whalesback Books.

Graburn, N. (1984). The Museum and the Visitor Experience. *Museum Education Anthology, 1973-1983: Perspectives on Informal Learning, a Decade of Roundtable Reports* (p. 177 – 182). Washington, D.C.: Museum Education Roundtable.

Kipper, G., & Rampolla, J. (2012). *Augmented Reality an emerging Technologies Guide to AR*. Waltham, MA: Syngress.

Lehn, D. v. (2006). Embodying Experience: A Video-based Examination Of Visitors' Conduct And Interaction In Museums. *European Journal of Marketing*, 40(11/12), 1340-1359.

Olofsson, U. K. (1979). *Museums and Children*. Paris: Unesco.